7 **(a)** A photon is a quantum (or packet/ discrete bundle) of electromagnetic radiation (energy) B1

Each quantum has energy equal to *hf* where *h* is Planck constant and *f* is the frequency of the electromagnetic radiation B1

(b) **(i)** **1.** $E_i = E_s + \frac{1}{2}mv^2$

2. $p_i = p_s \cos\theta + mv\cos\phi$

(ii) electron gain kinetic energy after collision, so scattered photon has lower energy than incident photon (refer to **(b)(i)1.**) B1

energy of photon inversely proportional to wavelength M1

so, wavelength of scattered photon larger than incident photon A1

(c) calculation for Δλ M1

λ_i / 10^{-12} m	λ_s /10^{-12} m	θ /°	Δλ / 10^{-12} m
191.92	193.27	57	1.35
153.30	154.65	57	1.35
965.04	966.84	75	1.8

valid conclusion comparing λ_i, θ and Δλ for the three sets of data e.g. A1

- compare first two sets of data, Δλ independent of incident photon wavelength for same scattering angle
- compare third set of data with the first two, Δλ dependent on incident photon wavelength for different scattering angles

(d) $\cos 70° = 0.342$, $\cos 75° = 0.259$, $\cos 80° = 0.174$ C1

either $\Delta(\cos\theta) = 0.342 - 0.259 = 0.08$
or $\Delta(\cos\theta) = 0.259 - 0.174 = 0.09$
or $\Delta(\cos\theta) = \frac{1}{2}(0.342 - 0.174) = 0.08$ C1

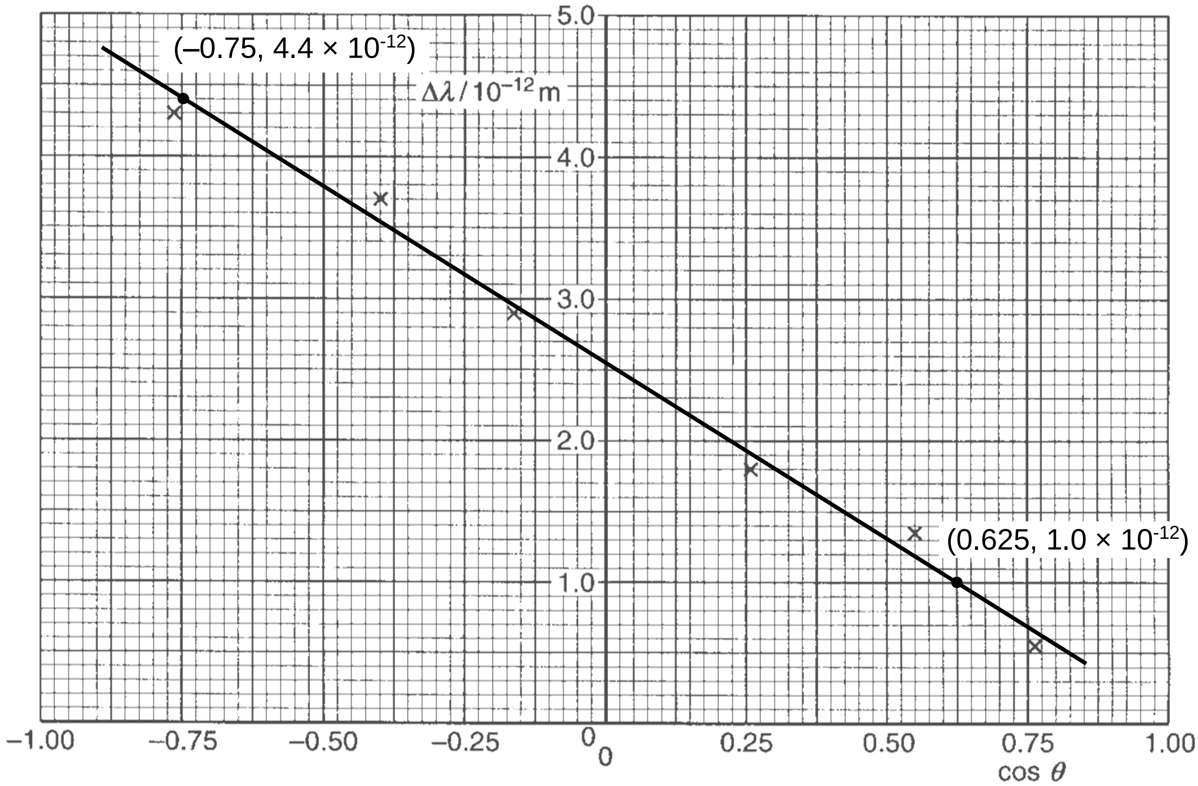
$\cos\theta = 0.26 \pm 0.08$ (or ± 0.09) A1

(e) **(i)** line of best fit assessed by even distribution of points either side of line along full length B1

(ii) [Method 1]
$\Delta\lambda = k(1 - \cos\theta) = -k\cos\theta + k$, so best fit straight line of Δλ against $\cos\theta$ gives M1

gradient equal –*k* and *y*-intercept equal *k* A1

[Method 2]
obtain values of $\Delta\lambda$ and $\cos\theta$ from a point on the best fit straight line and substitute into $\Delta\lambda = k(1 - \cos\theta)$ to calculate for *k*. B1



(iii) read-offs accurate to half a small square in both the *x* and *y* directions, gradient calculated correctly using $\frac{\Delta y}{\Delta x}$ and sign of gradient matches graph C1

$k = -\text{gradient}$ <u>and</u> with unit m A1

OR

$k = \text{y-intercept}$ <u>and</u> with unit m (M1)

precision of *y*-intercept to half the smallest square (A1)

OR

Substitute $\Delta\lambda$ and $\cos\theta$ from a point on the line into $\Delta\lambda = k(1 - \cos\theta)$ (C1)

k calculated correctly <u>and</u> with unit m (A1)

$(k \approx 2.5 \times 10^{-12}\ \text{m})$

(f) binding energy of electron less than / about $\left(\frac{10}{30000} \times 100\% =\right)$ 0.03% of energy of photon M1

photon energy >> binding energy so assumption of free electrons is valid / justified A1